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Raman study of shock features in plagioclase feldspar from the Mistastin Lake impact structure, Canada

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Abstract–Plagioclase feldspar is one of the most abundant minerals on the surface of the Earth, the Moon, and Mars, and is also commonly found in meteorites. Studying shock effects in feldspar thus provides us with fundamental information about impact cratering processes on planetary bodies. In this study, plagioclase from monomict and polymict breccias, impact melt rocks, and shock-metamorphosed target rocks, from throughout the Mistastin Lake impact structure, Canada, was examined using 514 nm laser Raman spectroscopy. As one of the very few impact structures with anorthosite in the target rocks, the Mistastin Lake impact structure provides a unique opportunity to study shocked plagioclase displaying progressive shock metamorphic features. A series of microscopic features was observed within plagioclase, including twins, needle-like inclusions, planar features, and alteration. The lack of planar deformation features is notable. Raman spectra of these features suggest that this technique is capable of differentiating and classifying shock features in low to moderately shocked rocks. Caution should be exercised, however, as Raman spectra collected from unshocked plagioclase references with known compositions indicate that peak width and peak ratio of the Raman peaks in lower wave number region (<350 cm^{-1}) and the main signature peaks around 500 cm^{-1} vary with chemical composition and crystal orientation. Data collected from diaplectic glass suggest that Raman features are efficient in distinguishing crystalline plagioclase and diaplectic glass. We also observed significant variations in the Raman intensities collected from diaplectic glass, which we ascribe to the localized disorder or inhomogeneity of shock pressure and temperature throughout the target.

INTRODUCTION

Impact cratering is a fundamental process that plays an important role in the formation and evolution of planetary bodies in the solar system. Minerals and rocks on planetary surfaces and interiors subjected to hypervelocity impact undergo a progression of deformation and display a variety of so-called shock effects (e.g., French and Short 1968; Stöffler 1972; Melosh 1989; French and Koeberl 2010; Ferrière and Osinski 2013). For the past half-century, shock effects have been studied extensively in rock-forming silicates, such as quartz and feldspar (e.g., French and Short 1968; Stöffler 1971; Robertson and Grieve 1977; Alexopoulos et al. 1988; Grieve et al. 1996). Nowadays, shock features in quartz are used as a standard diagnostic indicator for shock classification schemes, primarily due to its simple composition and crystal structure, together with a range of different shock deformation effects over a wide range of shock pressures, including planar deformation features (PDFs) and diaplectic glass (e.g., Robertson and Grieve 1977; Stöffler and Langenhorst 1994; Dressler 1990; Ferrière et al. 2009; French and Koeberl 2010; Stöffler et al. 2018; Pati et al. 2019). However, knowledge of shock metamorphism in quartz-free or quartz-poor, feldspar-rich systems, such as the Moon, Mars, and asteroids, is still poorly understood. Due to the complex chemical and crystal structure, the presence of solid solution, and the common presence of cleavage, it is more complicated to study shock metamorphism in feldspar

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than in quartz. Studies of shock features in feldspar were first conducted with optical microscopy using natural samples from terrestrial impact craters and meteorites (e.g., Chao 1968; Engelhardt and Stoffler 1968; Kieffer et al. 1976; Walawender 1977; Short and Gold 1996). A range of shock features were reported, including reduced refractive index, offset and bent twins, planar features, PDFs, mosaicism, partial to complete amorphization, and melting. Shock level classification systems based on these optical shock features were then developed (e.g., Stöffler 1971; Stöffler et al. 1991, 2018; Singleton et al. 2011).

To help understand the natural impact process, dynamic shock experiments and high-pressure static experiments on feldspar have also been conducted to reproduce shock features under controlled pressure and temperature conditions (e.g., Milton and De Carli 1963; Hörz and Quaide 1973; Ostertag 1983; Stöffler 1984; Tomioka et al. 2010; Jaret et al. 2018; Fritz et al. 2019). More recently, a number of workers employed advanced modern analytical techniques to study crystallinity changes and structural deformation in feldspar, including X-ray diffraction (XRD; e.g., Sims et al. 2019), cathodoluminescence (e.g., Kayama et al. 2009), thermal infrared (IR; e.g., Johnson et al. 2002, 2003), electron backscatter diffraction, micro-Fourier transform-infrared (micro-FT-IR) spectroscopy, and Raman spectroscopy (Velde and Boyer 1985; Velde et al. 1989; Heymann and Hörz 1990; Reynard et al. 1999; Fritz et al. 2005). Optical observations of experimentally shocked samples combined with spectroscopy data indicate that feldspar will progressively lose crystallinity and become increasingly amorphized with increasing pressure, eventually becoming completely amorphized (i.e., be converted to diaplectic glass). Among the various spectroscopy methods being employed, Raman spectroscopy has been widely used and has proven to be efficient for distinguishing and characterizing shocked minerals (Velde and Boyer 1985; Heymann and Hörz 1990; Reynard et al. 1999; Chen and El Goresy 2000; Farrell-Turner et al. 2005; Fritz et al. 2005, 2019; Freeman et al. 2008; Jaret et al. 2014, 2018). Raman spectroscopy is sensitive to chemical bond vibrations and crystal structure under different pressure–temperature conditions. For Raman features of silicates, the tectosilicate structure of feldspar with fully linked tetrahedra is distinct from those of ortho-, chain, ring, and layer silicates, in which the TO_4 tetrahedra are not linked with other TO_4 units (Deer et al. 2013). The Raman features of feldspar minerals are thus distinctly different from those of other tectosilicates such as quartz and zeolites (Sharma et al. 1983; Matson et al. 1986). Raman results from experimentally shocked feldspar show systematic changes including band broadening, reduction of intensity, and peak loss, as shock pressure increases (e.g., Jaret et al. 2018; Fritz et al. 2019). However, there have been no systematic Raman spectroscopy studies of naturally shocked feldspar from a complete range of shock levels from a specific terrestrial impact structure. Due to a combination of erosion, poor exposure, and/or sampling limitations, previous studies have mostly focused on feldspar in clasts in impact melt rocks (Grieve 1975; Marchand and Crocket 1977; McCormick et al. 1989; Marion 2009; Marion and Sylvester 2010; Jaret et al. 2014; Mader and Osinski 2018) or specifically on the formation of diaplectic glass in the case of plagioclase compositions, also known as maskelynite (Bunch et al. 1967; Engelhardt and Stöffler 1968; Hörz and Quaide 1973; Stöffler et al. 1991; Chen and El Goresy 2000; Rubin 2015; Jaret et al. 2015). This motivated our Raman study of plagioclase feldspar from the Mistastin Lake impact structure, Canada.

The Mistastin Lake impact structure is located in central Labrador, Canada (55°53′N; 63°18′W). It was dated at 36.6 ± 2.0 Ma by Young et al. (2013) based on in situ laser ablation $^{40}Ar/^{39}Ar$ geochronology dating on impact melt rock samples. The structure comprises an oval-shaped lake located in a large depression that is enclosed by a ring of hills approximately 28 km in diameter that are generally regarded as the apparent crater rim (Grieve 2006). The target rocks are dominated by granodiorite, quartz monzonite, and anorthosite (Grieve 1975; Mader and Osinski 2018), providing an excellent opportunity to better understand how impact events modify crystalline target rocks and minerals. The majority of studies of the Mistastin Lake impact structure have focused on geochemical analysis and characterization of the various impact breccias and impact melt rocks (Currie 1971; Grieve 1975; Marchand and Crocket 1977; Velde and Boyer 1985; Marion and Sylvester 2010; Mader and Osinski 2018). Pickersgill et al. (2015a, 2015b) examined shock features in plagioclase feldspar from the Mistastin Lake structure using optical microscopy and XRD. It is notable that with an X-ray beam size of 300 μm, phase identification from XRD measurements is often poorly constrained. Our study uses Raman spectroscopy with a laser beam size of 1–2 μm, which permitted us to examine finer features within shocked plagioclase feldspar. In addition, we also collected Raman spectra from different orientations of unshocked plagioclase feldspar crystals to provide orientation-dependent Raman spectra references. This work is also motivated by the fact that the Raman spectroscopy is a non-destructive technique, sensitive to crystallinity and deformation. Raman spectrometers will fly on both the Mars 2020

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Perseverance rover and the ExoMars 2022 rover missions (e.g., Beegle et al. 2015; Rull et al. 2017). Given the high probability that both these missions will encounter shocked material on the surface of Mars, our work may serve as a fundamental Raman reference for these upcoming missions.

METHODS AND SAMPLES

Shocked feldspar-bearing rock samples were collected during three field seasons (2009, 2010, and 2011) from a range of locations throughout the Mistastin Lake impact structure (Fig. 1). Sample locations and sample details are listed in Table 1. Most samples in this study were collected from anorthosite target rock, monomict breccia, and melt-free, -poor, and -bearing impact breccias, overlapping with most of the samples used in the earlier studies of Pickersgill et al. (2015a, 2015b) and Mader and Osinski (2018). In addition, four plagioclase crystals with different composition collected from Japan, Canada, Norway, and Virginia (see Table 2) were used as "unshocked" plagioclase reference material for Raman measurements.

Sixty-five polished thin sections were examined for microscopic features, using a Nikon Eclipse LV100POL compound petrographic microscope. Quantitative chemical composition was determined using a JXA-8530F field emission electron probe microanalyzer (FE-EPMA) in the Earth and Planetary Materials Analysis Laboratory at the University of Western Ontario. Beam operating conditions included accelerating voltage of 10–15 kV, probe current of 20 nA, and a beam diameter of ~5 μm. Mineral calibration standards used for wavelength dispersive spectrometry analyses were as follows: Albite (CM Taylor) for Na and Si, Orthoclase (CM Taylor) for Si and K, and Anorthite (Smithsonian USNM 137041—Great Sitkin Island, AL) for Al and Ca. The detection limit for these elements is 100 ppm. To observe the compositional variation, data from several spots within one crystal/target area were collected. To prevent Na loss caused by ion loss during electron beam bombardment, measures as suggested in Reed (2005), including reducing beam current and defocussing the beam, were taken into account.

Five plagioclase samples were powdered and their crystal structure was determined by powder XRD using a Bruker D8 Discover diffractometer with theta-theta instrument geometry. The geometry has a sealed cobalt source, Gobel mirror parallel beam optics, a pinhole collimator (100 or 300 μm), and two-dimensional (2D) general area detector diffraction system. The resulting diffraction patterns were imported into the evaluation software (Bruker DiffracPlus™ EVA) where phases were identified using the International Center for Diffraction Data database after baseline subtraction. Rietveld refinement was also conducted to calculate stress and strain.

Raman spectra of microscopic features in 65 thin sections were collected using a Renishaw InVia Reflex Raman Spectrometer at Surface Science Western. An argon-ion laser with a wavelength of 514 nm was used as exciting source. The spectrometer was calibrated by a Si film. The sample excitation and Raman signal collection were performed using 50× (numerical aperture [N.A.] = 0.75) and 100× (N.A. = 0.90) objective lenses in confocal mode on the Raman microscope. The laser spot size was focused to ~2 μm on the sample with the energy no more than 6 mW. The Raman spectra were collected from 126 to 2000 cm^{-1} with 1800 grooves per mm grating in a confocal mode. The spatial resolution is 0.5 cm^{-1}, the full widths at half maxima (FWHM) resolution is 0.3 cm^{-1}. The collecting time varied from 60 to 720 s in order to obtain sufficient intensity. The Raman shift of each band was obtained by Lorentzian curve fitting using the PeakFit program (SPSS Inc., Chicago). To make sure the results were consistent, 4–10 spectra within each target area were collected, and the spectra presented here are representative of the designated target. Importantly, all spectra presented in the same figure were collected with the same settings and collection time. In addition, to explore the influence of epoxy from the thin section, Raman spectra of 20 rock slabs (leftover from making the thin sections) were also collected. Raman spectra of unshocked plagioclase crystals with different composition were also measured as references.

RESULTS

Chemical Composition of Feldspar

The electron microprobe data indicate that the feldspar samples from Mistastin Lake are dominated by plagioclase feldspar with compositions in the range anorthite 28–55% (Fig. 2). Most of the plagioclase crystals from the faulted rim region and from the central part of the impact structure have a composition around An_{50} with a low K content ($Or_{1\text{-}6}$), whereas some plagioclase grains collected near the rim have relatively lower An content around 30 (Fig. 2). Backscattered electron (BSE) images of most plagioclase grains (Fig. 3) do not show zoning, but variable amounts of small K-feldspar (composition of Or_{94}) exsolution lamellae are present. No direct relation between the exsolution lamellae and shock features were observed. We selected four unshocked plagioclase specimens for analysis, as listed in Table 2, together

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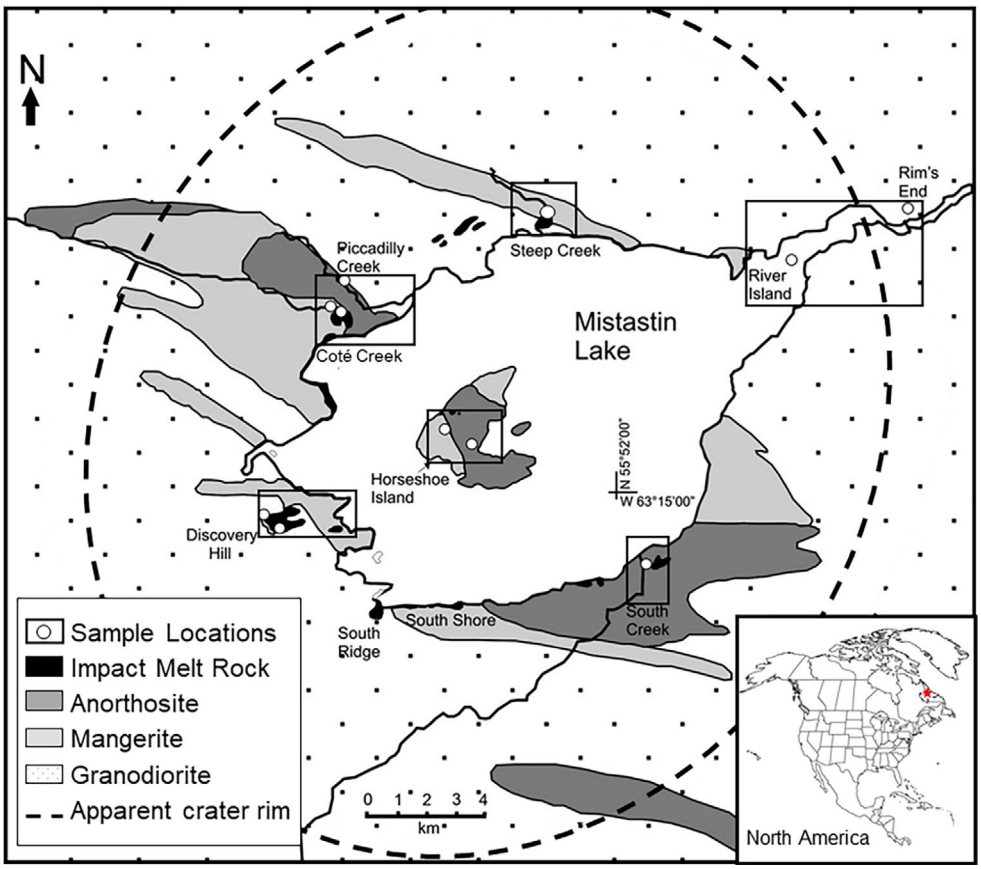


Fig. 1. A simplified geological map of the Mistastin Lake impact structure showing three main target lithologies (anorthosite, granodiorite, and mangerite). Boxes outline areas of interest and the locations of the collected samples are marked on the map as open circles. Note that outcrops of the various impact breccias are too small to be shown at this scale. The dashed line indicates the apparent crater rim adapted from Mader and Osinski (2018). Modified from Marion and Sylvester (2010) and Pickersgill et al. (2015a). (Color figure can be viewed at wileyonlinelibrary.com.)

with the average composition of plagioclase and diaplectic plagioclase glass from Mistastin Lake.

Raman Spectra of Unshocked Plagioclase

The Raman spectra of the plagioclase reference materials are provided in Fig. 4, together with one spectrum from K-feldspar (Or_{94}, sample MHI10-17 from the central uplift in Mistastin Lake). The most characteristic Raman features of feldspars are observed to be two or three Raman peaks between 460 and 515 cm^{-1}; the exact position of these peaks and the distance between the two diagnostic peaks vary with composition. The composition and the FWHM of the indicative bands between 460 and 515 cm^{-1} are listed in Table 3. Peak positions of plagioclase references materials and vibrational band assignments are listed in Table 4 (Sharma et al. 1983; McKeown 2005). The Rietveld refinement method was applied and confirms these reference samples to be stress- and strain-free.

For anorthite, the two strongest bands correspond to motion of the oxygen atom along a line bisecting the T-O-T angle (T = Si or Al) and the peaks in the 400–600 cm^{-1} region are characteristic of the four-membered rings of tetrahedra in the feldspar structure (Sharma et al. [1983] and references within). Among plagioclase with other compositions, this symmetric stretch v_s(T-O-T) mode is also observed to be the strongest, but the intensity and positions of the major peaks are slightly different. Our data suggest that changes in cations and the Si/Al ratio in feldspar have limited influence on the Raman position of the v_s(T-O-T) band. It should also be noted that plagioclase of intermediate composition (e.g., labradorite) has fewer peaks than those of the endmembers (i.e., albite and anorthite), and the two distinct peak positions are not exactly at the midpoint between two crystalline endmembers. Instead, the peaks occur at higher frequencies and are slightly broader than albite and anorthite. The Raman spectrum of K-feldspar is

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Table 1. Sampling and sample information for the Mistastin Lake impact structure samples used in this study (modified from Pickersgill et al. 2015a) and the shock features observed in plagioclase feldspar.

Crater location	Geographic location	Rock type	Sample name	Shock features observed in plagioclase
Central uplift	Horseshoe Island	Parautochthonous target rock (anorthosite)	MHI10-04, 17, 23, 35, 51, 54, 54-2; MST09- 25, 26	Fractures, deformed twins, undulatory extinction, planar features, diaplectic glass
Inner terrace	Coté Creek	Clast-rich impact melt rock; monomict breccia; polymict impact breccia (melt bearing)	MM09-35A, 35D, 35E; MM10-05-B1, 05-B2, 06-A2, 06-A3, 06-D2, 09-B, 10, 11, 12-2, 13-1, 13-2, 16, 17-A, 17-B, 34-A, 34-C2, 34-C5	Fractures, deformed twins, undulatory extinction, diaplectic glass
	Piccadilly Creek	Monomict breccia	MM10-20, 20-1, 20-2, 24, 25, 28, 30	Fractures, deformed twins, undulatory extinction, diaplectic glass
	South Creek	Monomict breccia	MM10-36-B1, 36-B2, 38, 39, 40, 41-1, 42, 43, 44, 45, 46-2	Fractures, deformed twins, undulatory extinction, diaplectic glass
Inner terrace	Discovery Hill	Polymict impact breccia (melt-bearing)	MM10-01-C	Fractures, undulatory extinction
Terrace	Steep Creek	Polymict impact breccia (melt-bearing)	MM09-10	Fractures, undulatory extinction, diaplectic glass
Rim	River Island	Target rock (mangerite, granodiorite)	MM10-32-A, 32-B, 33	Fractures, undulatory extinction
	Rim's End	Target rock (mangerite)	MM10-47, 48	Fractures

Table 2. Composition of plagioclase and diaplectic glass from Mistastin Lake and unshocked plagioclase references materials, obtained by EPMA.

	Mistastin Lake		Mistastin Lake		Amelia Courthouse, Virginia	Risor, Norway	Labrador, Canada	Miyake-Jima, Tokyo, Japan
	Plagioclase		Diaplectic glass		Albite	Oligoclase	Labradorite	Anorthite
Location	wt%	SD	wt%	SD	wt%	wt%	wt%	wt%
SiO_2	55.90	0.65	55.82	0.62	69.35	66.64	56.23	44.57
Al_2O_3	27.59	0.5	28.18	0.4	19.65	21.58	27.99	35.49
Na_2O	5.21	0.33	4.92	0.5	11.51	10.21	5.25	0.45
CaO	10.40	0.4	10.20	0.4	0.00	2.36	10.47	19.15
K_2O	0.38	0.08	0.57	0.07	0.16	0.26	0.39	0.02
FeO	0.25	0.05	0.21	0.05	0.00	0.11	0.23	0.39
MgO	0.00	0.02	0.01	0.02	0.00	0.00	0.01	0.09
TiO_2	0.03	0.01	0.00	0.01	0.03	0.02	0.01	0.01
BaO	0.05	0.01	0.05	0.01	0.00	0.00	0.02	0.00
Total	99.80		99.96		100.71	101.17	100.58	100.17
An	51.27		51.58		0.01	11.15	51.25	95.83
Ab	46.52		44.97		99.07	87.41	46.48	4.08
Or	2.20		3.45		0.92	1.44	2.26	0.09

wt% = mean composition in weight%; SD = standard deviation; An-Ab-Or content in mol%.

different from those of the plagioclase solid solution series with different peak positions and distances between the two/three most intense peaks. Raman data can, therefore, be used to distinguish K-feldspar exsolution in small areas (e.g., Fig. 3B) without the need for SEM analysis or EPMA data.

The second most intense set of Raman peaks are the ones below 350 cm^{-1}. The peaks in this range are reported to be from lattice vibration (Wood 1964; Sharma et al. 1983; McKeown 2005). Raman spectra from different crystal faces of the single crystal plagioclase An_{51} reference were collected and stacked in

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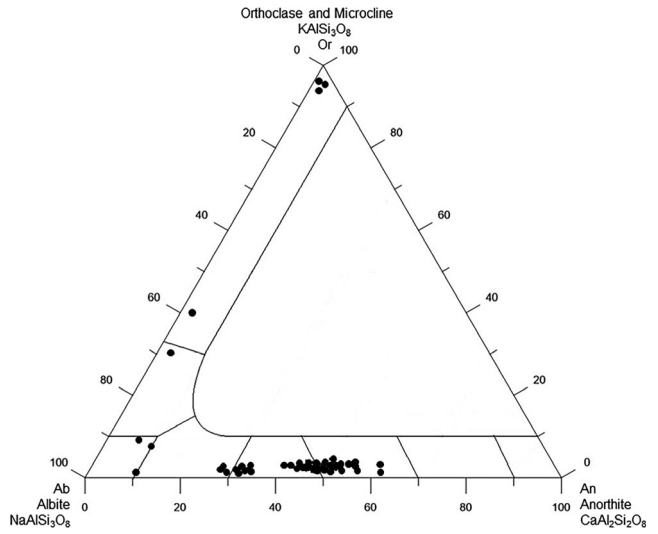


Fig. 2. Ternary diagram showing the chemical composition of feldspar from the Mistastin Lake impact structure.

Fig. 5. Importantly, peaks positioned below 350 cm^{-1} show similar wave numbers but vary in intensity, indicating an orientation-dependent intensity on the lattice vibration. The intensity ratio of two signature peaks between 480 and 510 cm^{-1} also varies with different crystal orientation (Fig. 5).

Raman peaks above 600 cm^{-1} tend to be relatively weak and their positions are all depending on the composition. According to Sharma et al. (1983), the high wave number modes in the 900–1200 cm^{-1} region that give rise to relatively weak Raman bands but strong IR bands, are associated with the motion of the oxygen atom along the line parallel to T-T and are assigned to the asymmetric stretch v_{as}(T-O-T) mode.

Raman Spectra of Microscopic Features in Shocked Plagioclase

A variety of shock features were observed within plagioclase feldspar samples from the Mistastin Lake impact structure, including fractures, deformed twins (curved and kinked twins), undulatory extinction (or undulose extinction), planar features, and partially isotropic to fully isotropic plagioclase (i.e., diaplectic glass). PDFs were also documented in quartz. Whole rock impact glass is also present in some of the impact melt-bearing breccia samples, but is not a focus of this study. The locations of the impact breccia samples are provided in Fig. 1 and Table 1; a summary of the observed shock features is also provided in Table 1.

Fractures and Deformed Twins

Fractures and twinning are common in plagioclase; although twinning can only be observed under cross-polarized light. Deformed twins were observed in several samples from throughout the Mistastin Lake structure, including bent, kinked, and offset features (Fig. 6). Some plagioclase containing kinked twins displays more fractures; more fractures do not necessarily mean more deformed twins. Raman spectra collected from heavily fractured areas in plagioclase grains demonstrate almost indistinguishable Raman features from the ones of unshocked areas. For the spectra collected from undeformed twins (e.g., spectra A and B in Fig. 7), the two most intense diagnostic peaks stay in the same position at 480 and 510 cm^{-1}, while the peaks from lattice modes below 350 cm^{-1} show slightly different features, which can be attributed to the different crystal orientations. Comparing the spectra of deformed (spectra C and D in Fig. 7) and undeformed twins (spectra A and B in Fig. 7), the Raman features collected from the most distorted areas show weaker intensity at lower wave number peaks. For the deformed twins, the weaker intensity of Raman peaks of lattice mode under 350 cm^{-1} suggests slight deformation of the crystal lattice.

Planar Features

While fractures and deformed twins may not be always shock related, PDFs in quartz have been widely accepted as diagnostic indicators of shock metamorphism (e.g., French and Koeberl 2010). Given well-developed, undecorated PDFs in quartz have been observed in samples from the central uplift of the Mistastin Lake impact structure (Taylor and Dence 1969; Pickersgill et al. 2015b), we conducted a thorough search for PDFs in plagioclase. Consistent with previous reports (Pickersgill et al. 2015b), we observed no PDFs; however, we did observe a few microscopic planar features and they are described below.

Needle-Like Inclusions

Under transmitted light, dark needle-like orientated inclusions were observed within most unshocked and shocked plagioclase crystals and diaplectic glass from samples taken throughout the crater, but not in the impact melt rock samples (Figs. 8A and 8B). The needle-like inclusions are dark red to black in color with a thickness <5 μm. Raman spectra confirm that these needles are hematite and/or magnetite (Fig. 8C), consistent with the previous suggestion of these features being "iron-oxide" by Pickersgill et al. (2015b). Their presence in both unshocked crystalline plagioclase and diaplectic glass indicates that these inclusions existed prior to the impact event and survived the formation of diaplectic glass.

Alteration

A fine fibrous texture (each fiber is ~1 μm wide) was observed in plagioclase from the central uplift of

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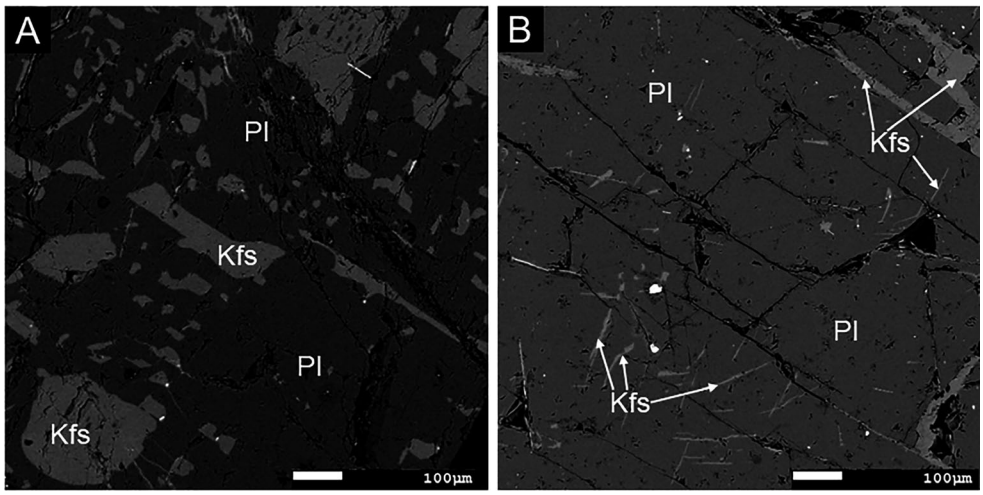


Fig. 3. Backscattered electron images showing K-feldspar (Kfs, lighter gray) exsolution within plagioclase grains (Pl, darker gray) from the central uplift (Horseshoe Island) in Mistastin Lake. A) K-feldspar patches in plagioclase (sample MHI10-23) and (B) fine K-feldspar exsolution lamellae (sample MHI10-17).

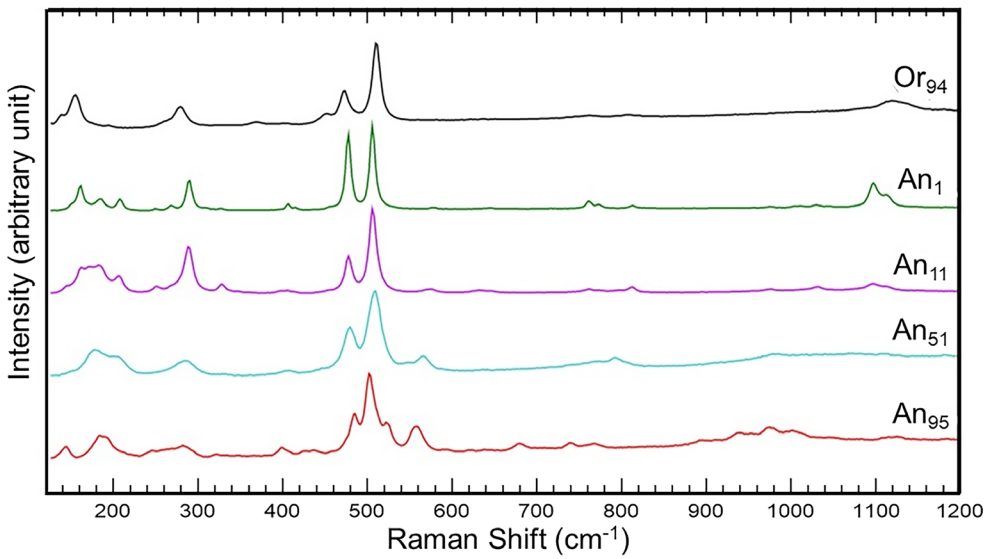


Fig. 4. Raman spectra of plagioclase reference materials with different chemical compositions. (Color figure can be viewed at wileyonlinelibrary.com.)

the Mistastin Lake structure (Figs. 9A and 9B). Our observations of these fibrous features suggest that they mainly occur within certain twins of shocked plagioclase. Raman spectra were collected on the fine fibers as well as the surrounding area (Fig. 9C). Previous XRD and EPMA results suggested that this fibrous phase was levyne-Ca zeolite (Pickersgill et al. 2015b). Raman spectra from the fibrous area show a peak around 500 cm^{-1} and a broad band centered around 3550 cm^{-1}. Comparing with the RRUFF database and previous Raman studies (e.g., Knops-Gerrits et al. 1997), our Raman results confirm this phase as a zeolite, and the peak around 500 cm^{-1} is from ν_s(T-O-T; T = Al, Si) and the broad band centered around 3550 cm^{-1} is from the OH stretching mode. Raman spectra were collected from various locations within the same plagioclase grain. From the plagioclase twin without alteration (spots C and D in Fig. 9) to the alteration fibers (spot A in Fig. 9), the two main Raman peaks near 500 cm^{-1} gradually merge together to form one peak and the OH band centered around 3550 cm^{-1} appears and intensity increases. Raman results suggest that the twin containing the fibrous area bears more water than the unaltered twin which shows no OH band in the 3000–4000 cm^{-1} range, consistent with the presence of zeolite alteration.

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Table 3. Raman peak positions and FWHM of different plagioclase compositions from this study.

Plagioclase	Composition	Main Peak (cm^{-1})		FWHM (cm^{-1})
Anorthite	$An_{96}Ab_3Or_1$	<	485.2	11.0 (5)
			503.9	13.5 (5)
Labradorite	$An_{51}Ab_{46}Or_3$	<	479.6	17.0 (5)
			509.8	19.8 (5)
Oligoclase	$An_{11}Ab_{87}Or_2$	<	478.5	9.6 (5)
			507.3	10.7 (5)
Albite	$An_1Ab_{98}Or_1$	<	478.4	7.6 (5)
			506.8	7.9 (5)

Planar Features

In this study, we observed some planar features in plagioclase in several samples, with the greatest abundance in shocked target rocks from the central uplift (Fig. 10). These planar features always occur in samples in which quartz with PDFs is present. The planar features observed in plagioclase occur mostly near the edge of two crystals and are 2–4 μm thick (Figs. 10A and 10B). They are not open fractures when examined under reflected light. Compared to the mechanical twins observed throughout the crystal (Figs. 10C and 10D), the planar features are narrower and are darker under cross-polarized light (Figs. 10E and 10F). Raman spectra of the surrounding birefringent area exhibit normal Raman features from plagioclase (spectra C and F in Fig. 10G); whereas spectra collected from the planar features (spectra A and D in Fig. 10G) exhibit the two main Raman peaks but they are broader. The same position of these Raman peaks indicates that they share the same composition and are not K-feldspar exsolution lamellae. The observable Raman peaks suggest that the planar features are not amorphous. However, when compared to the surrounding birefringent area, the Raman spectra collected within the planar features show broader peaks and loss of weak peaks below 350 cm^{-1}. The loss of peaks from lattice modes indicates a much more disturbed crystal structure, while the broadening of the two main peaks suggests deformation of the TO_4 tetrahedra.

Diaplectic Glass

Petrographically, diaplectic glass appears identical to plagioclase in plane-polarized light, preserving crystal shapes and boundaries such that the texture of the crystal itself and the rock remains. In cross-polarized light, diaplectic glass is easily distinguishable, as it is optically isotropic. Among the samples from Mistastin Lake, diaplectic glass is mostly observed in the melt-poor breccia samples from the inner terrace and shocked anorthosite samples from the central uplift.

Table 4. Raman frequencies (cm^{-1}) for members of the Ab–An plagioclase series collected in this study and vibrational band assignments.

An01	An11	An51	An95	Anorthite (Sharma et al. 1983)	Albite (McKeown 2005)
–	–	–	144	Cation and lattice mode	Tetrahedral cage-shear modes
162	162	178	–		
185	184	–	184		
209	207	204	–		The four-membered ring rotation–translation modes
251	251	–	247		
269	–	–	–		
290	290	286	283		
327	328	–	322		
–	–	–	400		
407	407	–	–		
415	–	–	–		
–	–	–	426	ν_s(T-O-T)	Four-member ring-breathing modes of the tetrahedra
–	–	–	438	Symmetric	
–	–	–	–	T–O–T	
478	478	480	486	stretching	
507	507	519	504	modes	
–	–	–	525	(T: Si or Al)	
–	–	549	–		
579	576	567	559		
–	–	–	590		
–	–	–	621	Asymmetric	Tetrahedral deformation
–	633	–	639	O–T–O	
646	646	–	661	bending modes (T: Si or Al)	
–	–	–	742	T-T stretching	
763	763	–	769	modes (T: Si	
–	–	792	–	or Al)	
814	814	–	–		
–	–	–	893		
–	–	–	913	ν_{as}(Si-O-Al)	Vibrational stretching modes of the tetrahedra
–	–	–	940		
977	978	–	977		
–	–	987	–		
1008	–	–	1003		
1031	1033	–	–	ν_{as}(Si-O-Si)	
1044	–	–	–	T–O	
1100	1100	1084	–	stretching	
1112	–	–	1127	modes occurring	

These examples preserve the plagioclase crystal shape and texture (including inclusions) and are optically isotropic under cross-polarized light (Fig. 11). Most of the diaplectic glass observed at Mistastin Lake contains fractures and oriented needle-like inclusions described above, just like crystalline plagioclase. We analyzed 26 diaplectic glass grains, collecting 5–10 Raman spectra in each; three representative examples are shown in Fig. 12. These spectra were all collected using the same

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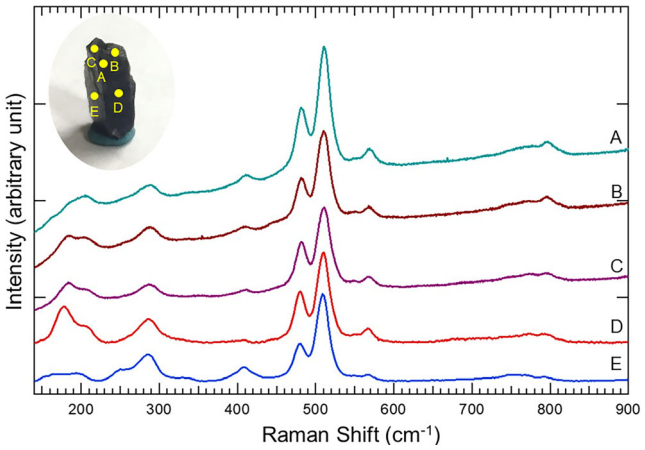


Fig. 5. Raman spectra collected from an An_{51} plagioclase single crystal at different orientations. Yellow dots indicate the crystal faces where Raman spectra were collected. (Color figure can be viewed at wileyonlinelibrary.com.)

Raman set-up and with the same collection time. Comparing the Raman features from unshocked plagioclase with diaplectic plagioclase glass, the two diagnostic peaks in the Raman spectrum of plagioclase at 480 and 509 cm^{-1} are no longer seen in the diaplectic glass spectra. Instead, there is a large hump centered at ~500 cm^{-1} with a broad shoulder at 580 cm^{-1}; all the other weak peaks are no longer seen, and another hump centered around 1000–1100 cm^{-1} is observed. The Raman peak positions of diaplectic glass with similar composition remain consistent; however, the intensity of the features is variable, in both rock slabs and thin sections. As seen in Fig. 12, the intensity of spectra A, B, and C is quite different: A shows the strongest intensity, while C shows the weakest. It is also notable that the intensity of the spectra collected across the same diaplectic glass grain can also show significant variation (e.g., Fig. 12A versus 12B, and Fig. 12B versus 12C). Similar intensity variation was still observable when we extended the collecting time from 120 s to 10 minutes.

DISCUSSION

Planar Features

Feldspars have been reported to display various shock-related planar structures, including planar fractures, deformation bands, mechanical twins, kink bands, and PDFs (e.g., French and Short 1968; Stöffler 1967, 1972; Stöffler and Hornemann 1972; White 1993; French 1998). However, except for PDFs, most of these planar structures are not uniquely indicative of shock and are not considered an unequivocal indicator of shock metamorphism (French and Koeberl 2010).

The most PDF-like features in plagioclase we observed were in a sample from the central uplift (Fig. 10). Under cross-polarized light, these planar features appear to be dark and isotropic while the surrounding area is birefringent. However, their Raman spectra still display features similar to crystalline plagioclase, suggesting that the planar features are not completely amorphous. Broadening of the main Raman peaks and loss of lattice mode bands in spectra from planar features suggest a more deformed TO_4 tetrahedra structure than surrounding birefringent areas (Fig. 10G). Thus, while these planar features we observed in plagioclase are not completely amorphous—as in PDFs in quartz—we suggest that these features are consistent with being formed via shock metamorphism.

As previously noted, the lack of observable PDFs in plagioclase is puzzling (Pickersgill et al. 2015b). There are three potential explanations for this conundrum: either PDFs did not form in plagioclase at Mistastin or they did form, but they are difficult to observe and/or were subsequently altered, also hindering their detection. It is possible that the development of planar features in feldspar is more complicated than in quartz. Indeed, shock experiments conducted at pressures required to create PDFs in quartz did not produce PDFs in shocked feldspar samples (e.g., Gibbons and Ahrens 1977; Jaret et al. 2018; Sims et al. 2019). This is consistent with our observations that PDFs are present in quartz but absent in plagioclase down to the thin section scale. One explanation is, therefore, that PDFs do not form in feldspar, yet as noted above, previous studies have documented the presence of such shock features. The simplest explanation is that the development of PDFs in feldspar is influenced by crystal structure and/or chemical composition. Unfortunately, many previous workers who documented the presence of PDFs in feldspar did not provide the composition, or even if the grain in question was plagioclase or K-feldspar. It is outside the scope of this contribution to review all previous determinations, but we do note that PDFs in previous reports appear to have been predominantly observed in K-feldspar (e.g., Ostertag 1983; Gibson and Reimold 2005; Nagy et al. 2008) and Na-rich plagioclase (e.g., Trepmann et al. 2003).

We suggest the previous report of PDFs in plagioclase at Mistastin Lake was likely due to misidentification (Taylor and Dence 1969). Indeed, we observed a variety of microscopic planar elements that may easily be mistaken for shock-induced planar features, including needle-like inclusions, alteration, and twins (Figs. 8–10). Micro-Raman spectra proved to be effective in discerning the origins of these planar

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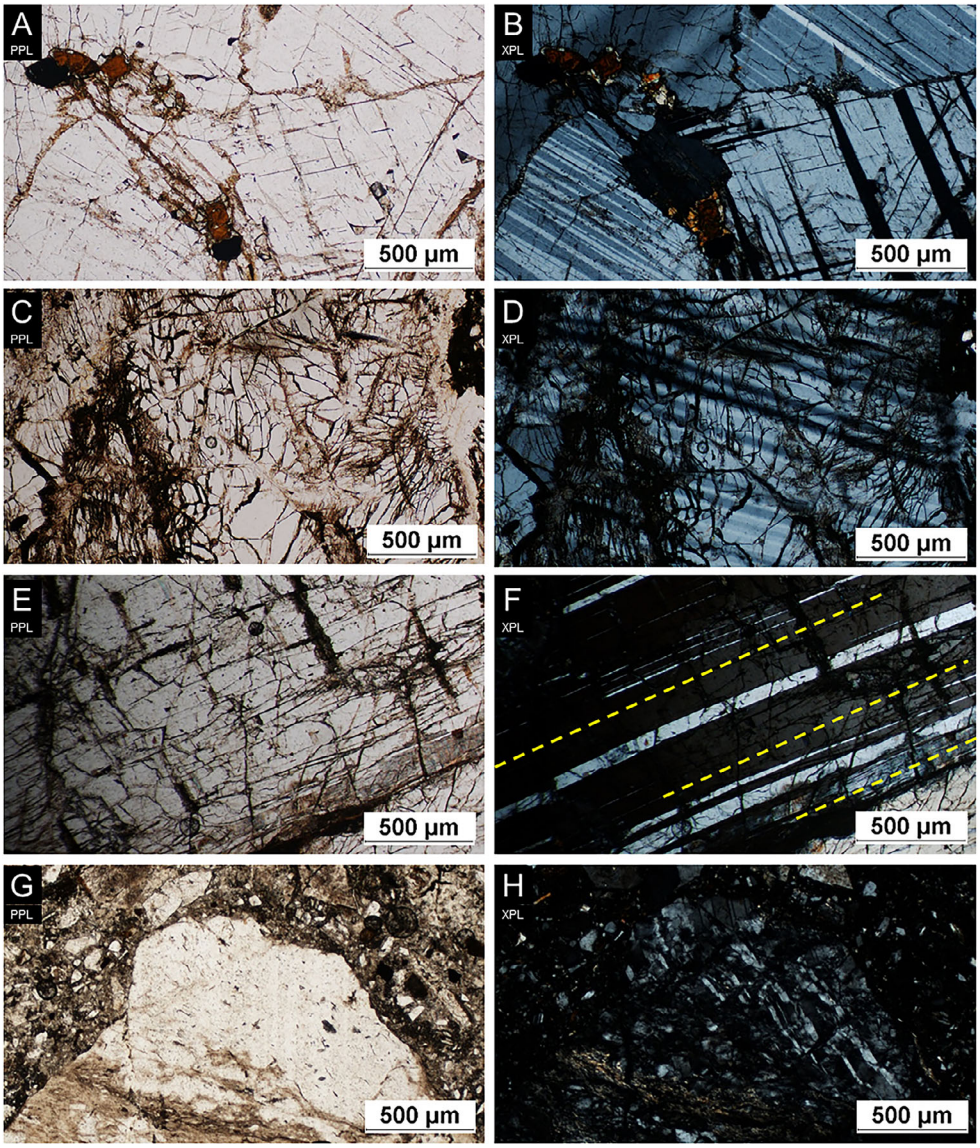


Fig. 6. Transmitted light photomicrographs of shocked plagioclase features. PPL = plane-polarized light; XPL = cross-polarized light. A, B) Fractures and cleavage are common within plagioclase from shocked anorthosite at Mistastin. Sample MM10-20-1 from the central uplift, Horseshoe Island. C, D) Irregular fractures within plagioclase (sample MM10-34C-5) from melt-poor polymict breccia, inner terrace, Coté Creek. E, F) Bent twins (seen under XPL, the twins are slightly curved compared to the straight dash lines) within plagioclase from shocked anorthosite, central uplift, Horseshoe Island (sample MHI10-54-2). G, H) Deformed twins with offsets and fractures forming (seen under XPL) within plagioclase from melt-poor polymict breccia, inner terrace, Coté Creek (sample MM10-34C-2). (Color figure can be viewed at wileyonlinelibrary.com.)

features. For the twins, Raman features below 350 cm^{-1} can be used as an indication of different crystal orientation (Fig. 7C). For inclusions, as they are usually other minerals with different chemical composition from feldspar, Raman spectra can be used effectively in telling them apart (Fig. 8C). For exsolution lamellae, K-feldspar can be identified based on the signature three peaks near 500 cm^{-1}, compared to the two peaks of plagioclase (Fig. 4). In terms of alteration, particularly the zeolite in this study, Raman features indicative of OH around 3500 cm^{-1} provide a strong indication of the presence of such OH-bearing alteration minerals (Figs. 9C and 9D).

The possibility that PDFs do form in plagioclase but that they become subsequently altered and destroyed remains a possibility. The gradual changes of Raman features from shocked plagioclase to water-bearing zeolite may suggest a preferential low-temperature hydrothermal levyne-Ca alteration of the shock-weakened plagioclase structure (Fig. 9). This is

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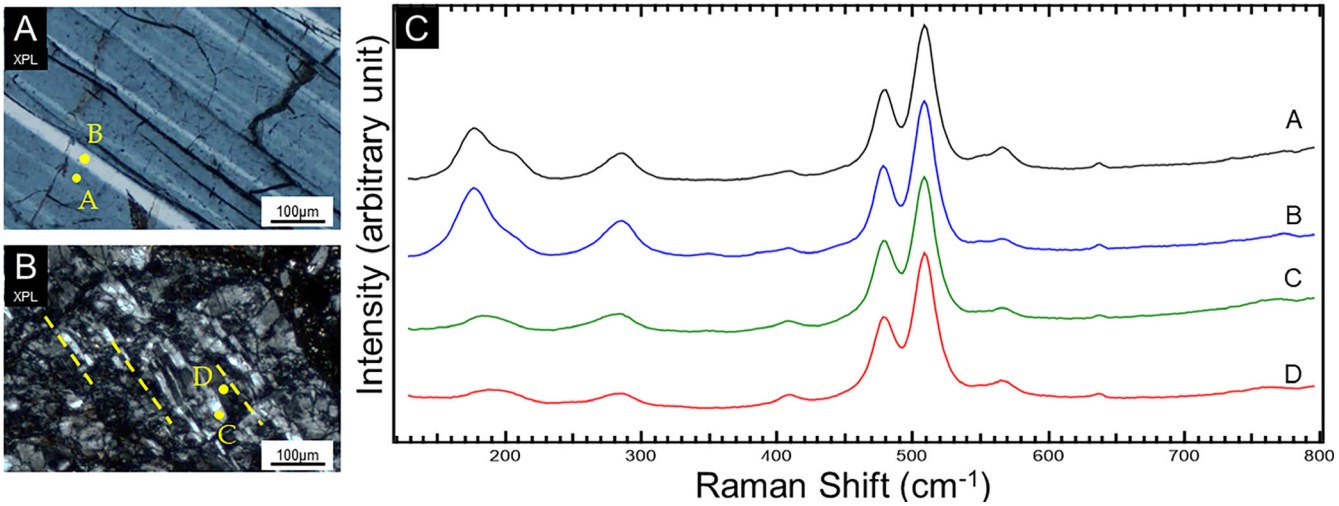


Fig. 7. Undeformed and deformed twins and their Raman spectra. XPL = cross-polarized light. Yellow dots indicate the location where the Raman spectra are collected. A) Undeformed twins. B) Deformed twins. The dashed line indicates the orientation of the twins. C) Raman spectra collected from the twins: spectra A and B are collected from spots A and B in image A, spectra C and D are collected from spots C and D in image (B). (Color figure can be viewed at wileyonlinelibrary.com.)

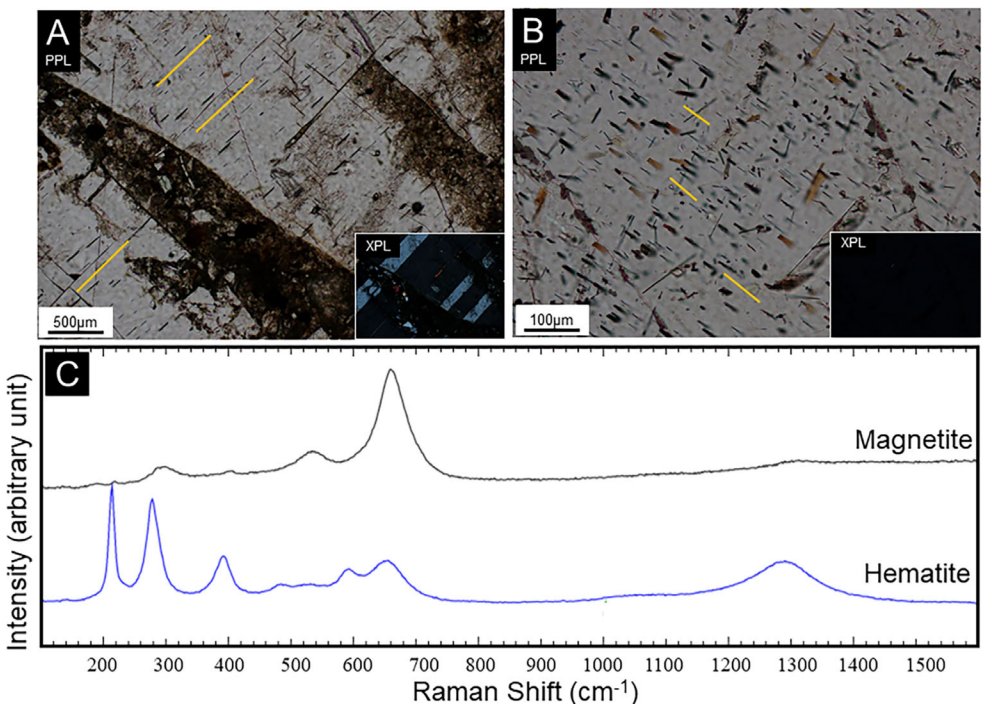


Fig. 8. Needle-like inclusions observed in plagioclase and diaplectic glass and their Raman spectra. Orientation of the needle-like inclusions is marked by yellow lines. The XPL image insets cover the exact same areas as the corresponding PPL images. PPL = plane-polarized light; XPL = cross-polarized light. A) Needle-like inclusions in plagioclase observed in sample MM10-34C-5 from melt-poor polymict breccia in Coté Creek. B) Needle-like inclusions in diaplectic glass observed in sample MST09-25 from anorthosite in the central uplift. C) Representative Raman spectra of the inclusions suggested them being a combination of magnetite and hematite. (Color figure can be viewed at wileyonlinelibrary.com.)

consistent with observations of feldspars from the Manson impact structure reported by Short and Gold (1996), who suggested that shock-damaged crystals are more susceptible to chemical breakdown and replacement than their unshocked equivalents. However, this does not explain why PDFs are lacking in

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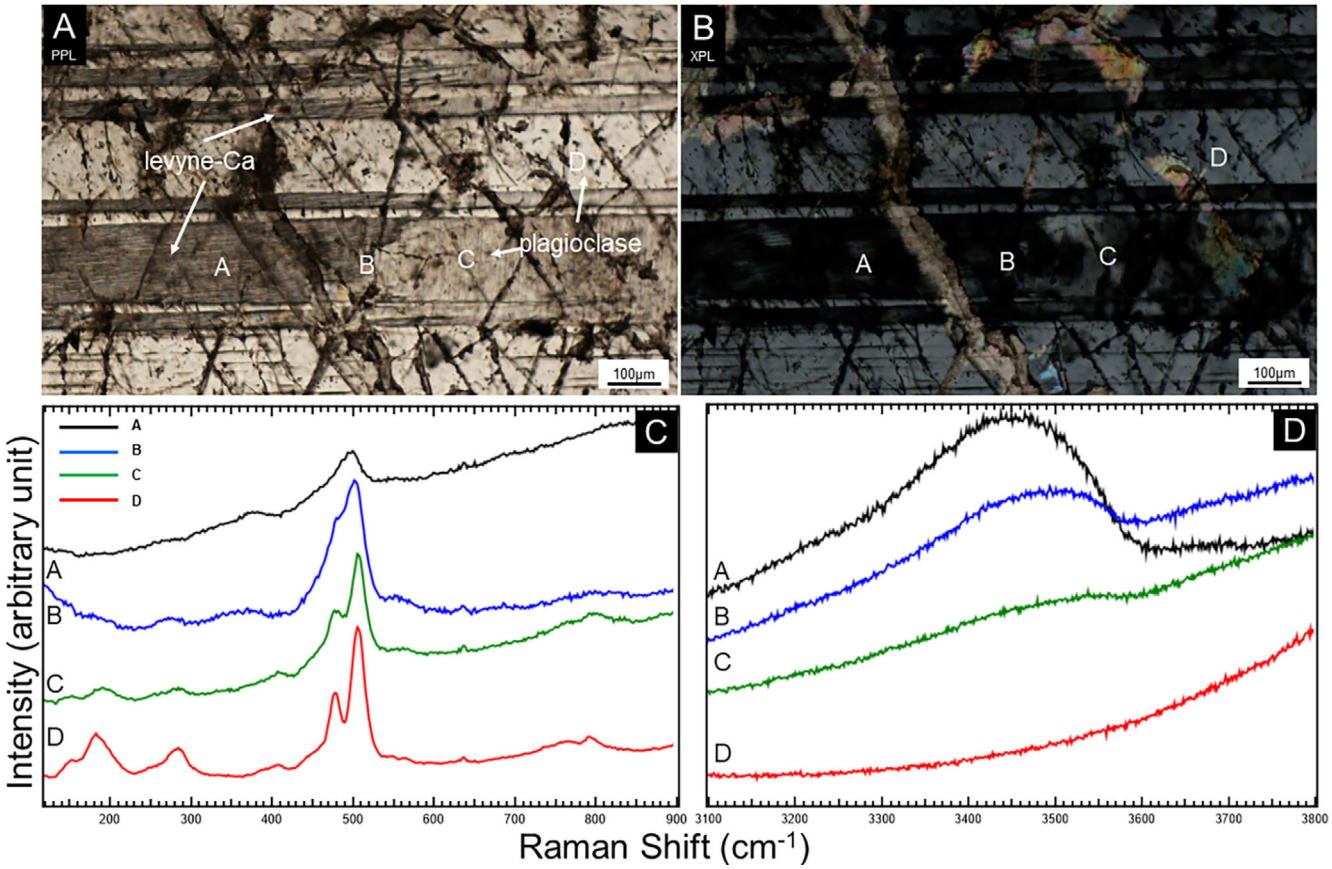


Fig. 9. Fibrous texture observed in sample MHI10-17 from shocked anorthosite (central uplift Horseshoe Island) and Raman spectra collected from various areas. A and B) Plane-polarized (PPL) and cross-polarized (XPL) photomicrographs of shocked plagioclase. Spot A is in the fibrous area, spot B is in the transition area, and spots C and D are from the unaltered plagioclase twins. C and D) Raman spectra collected from the spots labelled in (A) and (B). Spectrum C covers the 120–900 cm^{-1} range; spectrum D covers 3100–3800 cm^{-1} range. (Color figure can be viewed at wileyonlinelibrary.com.)

intermediate or Ca-rich plagioclase in contrast to Na-rich plagioclase and K-feldspar.

Diaplectic Glass

Diaplectic glass observed in this study preserves the original plagioclase crystal shape and the presence of needle-like inclusions, and is optically isotropic under cross-polarized light (Fig. 11). Raman spectra of diaplectic glass with intermediate plagioclase composition consist of a broad hump centered at ~500 cm^{-1} with a broad shoulder at 580 cm^{-1}, and another hump centered around 1050 cm^{-1}. Our Raman features of diaplectic glass are comparable to previously reported Raman spectra of diaplectic glass from both natural meteorite samples and experimentally shocked plagioclase (e.g., Velde et al. 1989; Daniel et al. 1997; Chen and El Goresy 2000; Fritz et al. 2005, 2019; Jaret et al. 2015, 2018; Pittarello et al. 2020; Sims et al. 2020).

When compared to crystalline plagioclase, the Raman features of diaplectic glass show some obvious changes: the two plagioclase-diagnostic peaks in the Raman spectrum merge to form a broad hump centered at ~500 cm^{-1}, while the peaks near 570 cm^{-1} broaden to form a shoulder around 580 cm^{-1}, and another broad peak emerges between 1000 and 1200 cm^{-1}. The positions of the main Raman peaks around 500 cm^{-1} in the diaplectic glass are closely related to those of the most prominent bands observed in the spectra of crystalline plagioclase, assigned to stretching and bending vibrations and TO_4 groups (Daniel et al. 1995). Sharma et al. (1983) suggested that the variation of the T-O-T angle is the basis of the short-range variability and long-range disorder and contributes to the broadening of the peak. Fritz et al. (2019) experimentally shocked intermediate composition plagioclase (An_{50-57}) and reported the intensity increase of the shoulder around 580 cm^{-1} from crystalline plagioclase to diaplectic glass. These authors suggested that this was due to an increase in the number of three-membered rings of tetrahedra at the expense of four-

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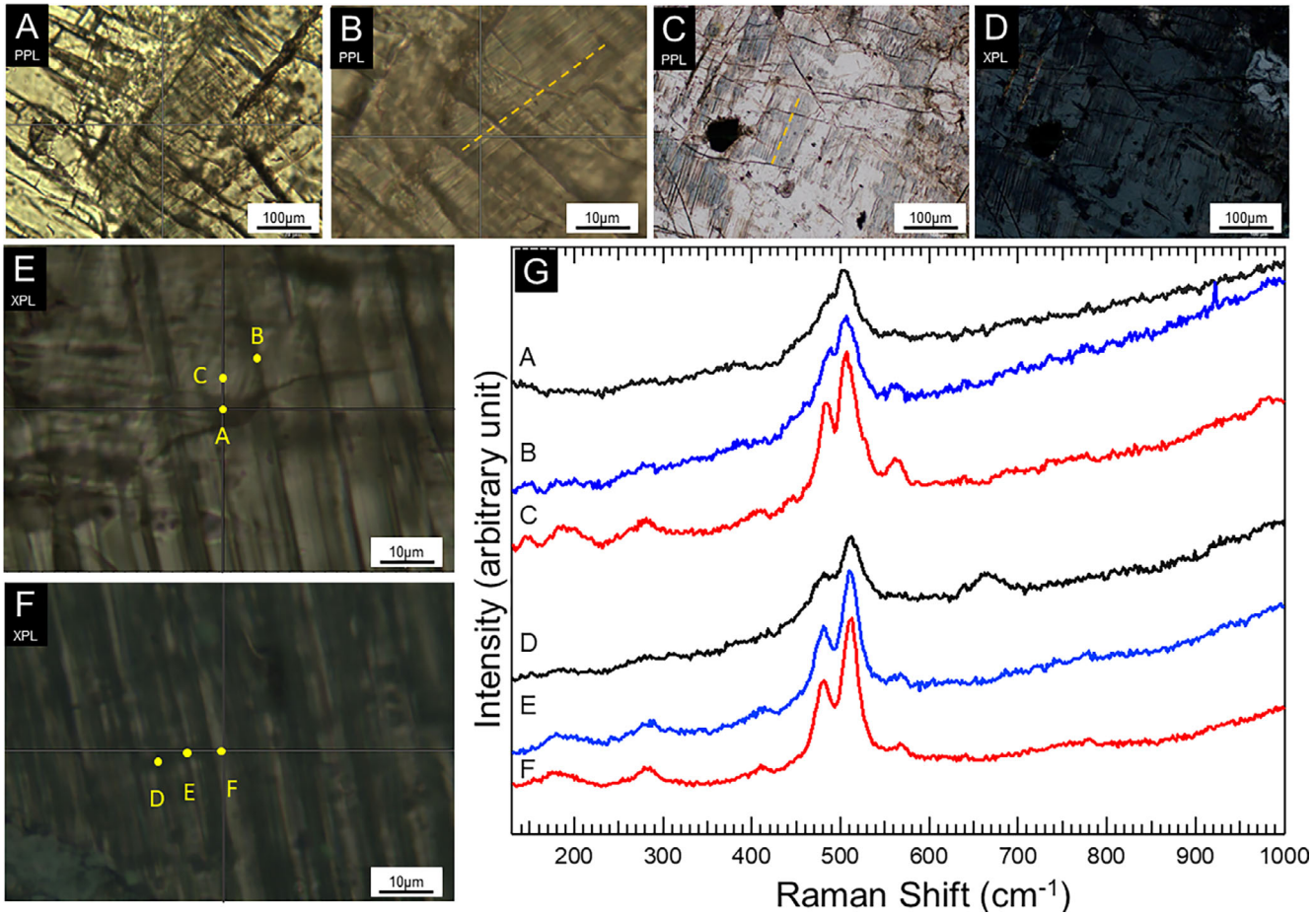


Fig. 10. Planar features observed in target rock anorthosite samples (MHI10-54) from the central uplift on Horseshoe Island. PPL = plane-polarized light; XPL = cross-polarized light. A, B) Planar features observed at the boundary of two plagioclase grains. C, D) Mechanical twins observed throughout the crystal. E and F) Close-ups of two areas with planar features. Yellow dots indicate where the Raman spectra were collected: spots A and D are directly on the planar features, spots B and E are in the transition area, and spots C and F are in the birefringent areas. G) Raman spectra collected from the spots in images (E) and (F) are labelled accordingly. (Color figure can be viewed at wileyonlinelibrary.com.)

membered rings. The broad peak centered around 1050 cm^{-1} is related to T–O stretching vibrations of nonbridging oxygen atoms attached to either Si or Al (Sharma et al. 1983; Matson et al. 1986). The absence of this broad peak in crystalline plagioclase can be explained by the lack of nonbridging oxygen-atoms in a well-crystallized tectosilicate (cf. Fritz et al. 2019). Deformation and breaking of tetrahedral bonds during progressive amorphization led to an increase in nonbridging oxygen atoms attached to Si/Al. Vibration of the nonbridging Si-O and Al-O units in a more or less random order contributes to the broad peak at 900–1150 cm^{-1}. Fritz et al. (2005) reported that the position of this broad peak is related to the diaplectic glass composition and the shift in position provides a measure of the Al-Si substitutions in the tetrahedron, thus reflecting the different Ca and Na distributions (Freeman et al. 2008).

We also observed a considerable variation in the intensity of Raman peaks from diaplectic glass with the same composition (Fig. 12). The diaplectic glass spectra with relatively higher intensity (e.g., Fig. 12A) are comparable with previous studies using natural samples, especially meteorites (e.g., Fritz et al. 2005). The spectra with weak intensities (e.g., Fig. 12C) are comparable with the ones reported using static experiments (e.g., Sims et al. 2020) and shock experiments (e.g., Jaret et al. 2018). Jaret et al. (2018) and Sims et al. (2020) compressed andesine (An_{43} and An_{49}, respectively) dynamically and statically to complete amorphization and the Raman spectra from the quenched samples showed weak broad bands in the low wave number (400–700 cm^{-1}) and no visible bands in the high wave number (900–1300 cm^{-1}) region. However, there has not been any previous study reporting the intensity variation of the broad peak centered around 1050 cm^{-1}. Consider this broad Raman

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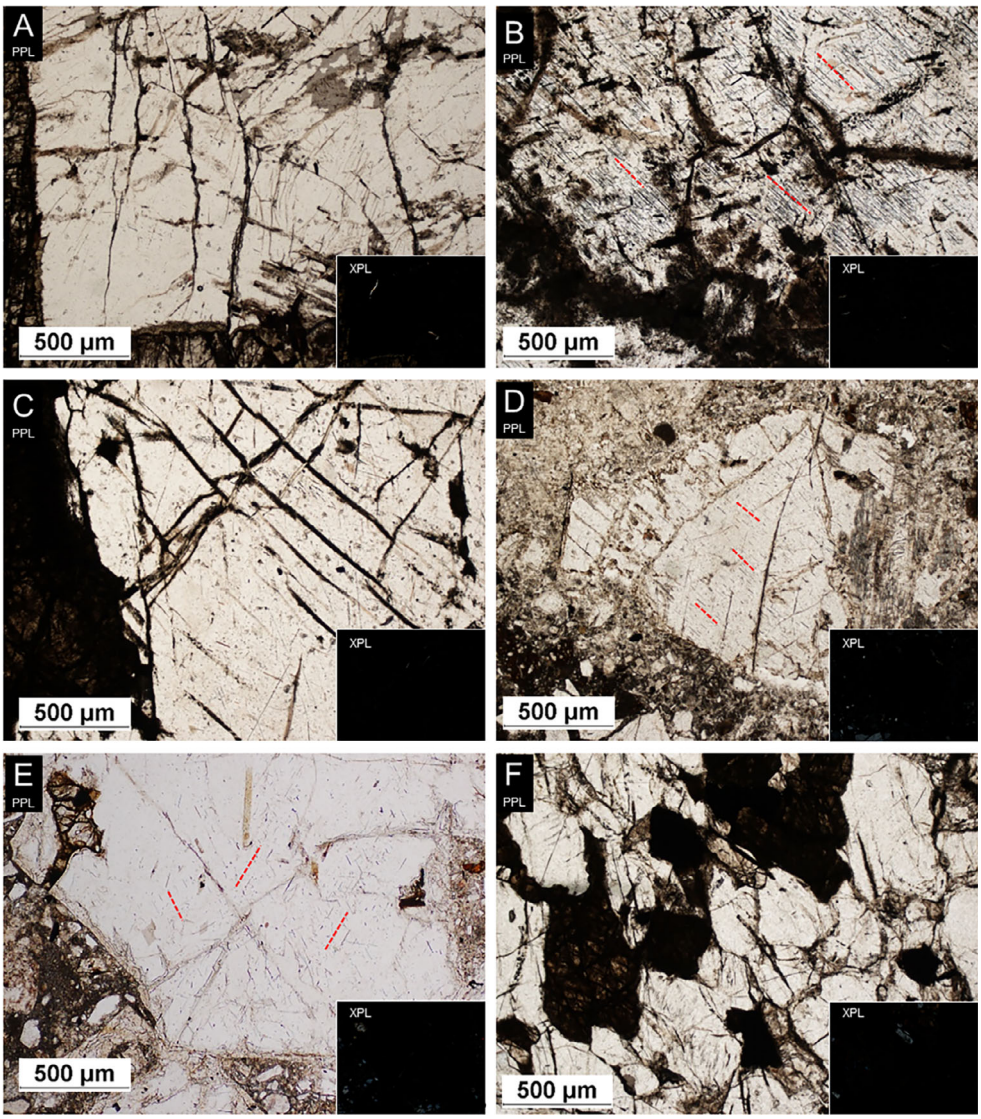


Fig. 11. Photomicrographs of diaplectic plagioclase glass from the Mistastin Lake impact structure. The needle-like inclusions are indicated by dashed lines. The XPL image insets are the exact same areas as the corresponding PPL images. PPL = plane-polarized light; XPL = cross-polarized light. A) Diaplectic glass in sample MM10-13 from melt-poor polymict breccia, Coté Creek. B) Diaplectic glass in sample MM10-38 from monomict breccia, South Creek. C) Diaplectic glass in sample MM10-34A from melt-poor polymict breccia, Coté Creek. D) Diaplectic glass in sample MM10-34C-2 from melt-poor polymict breccia, Coté Creek. E) Diaplectic glass in sample MM10-34C from melt-poor polymict breccia, Coté Creek. F) Diaplectic glass in sample MM10-34C-5 from melt-poor polymict breccia, Coté Creek (Color figure can be viewed at wileyonlinelibrary.com.)

peak comes from nonbridging oxygen atoms, increased numbers of nonbridging Si-O and Al-O from stronger shock pressure/temperature may contribute to the increased intensity.

We suggest that the intensity variation of Raman features from diaplectic glass is likely due to the localized short-range order of the inhomogeneous nature of diaplectic glass. It has been reported previously that diaplectic glass has a lower refractive index than unshocked plagioclase and that its refractive index decreases with increasing shock pressure and temperature (e.g., Stöffler 1972; Stöffler and Hornemann 1972; Fritz et al. 2019), suggesting that the formation of diaplectic glass might happen under a variety of pressure–temperature–time paths. Nevertheless, more future work is needed to fully understand the formation mechanism of diaplectic glass.

Shock Level Classification and Associated Raman Features

The principle of progressive shock metamorphism was first defined for quartzofeldspathic rocks by Stöffler

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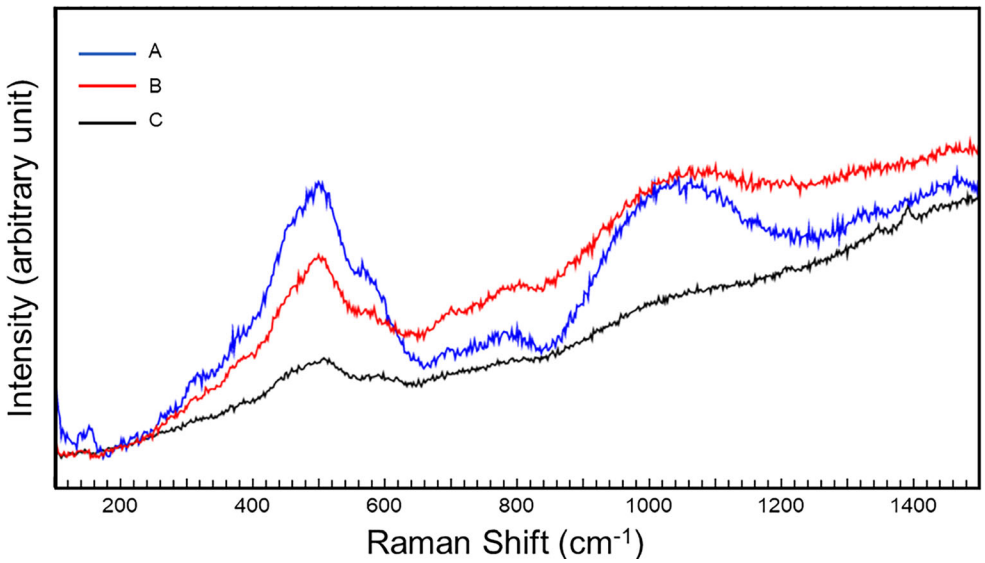


Fig. 12. Raman spectra of three different diaplectic glass samples: A (blue) is from Fig. 11A, B (red) is from Fig. 11B, C (black) is from Fig. 11C. All three spectra were collected with the same Raman setup and collecting time. Note that all three diaplectic glasses have the same composition An_{51}. (Color figure can be viewed at wileyonlinelibrary.com.)

(1965, 1966), and Chao (1967). Recently, Stöffler et al. (2018) proposed the most up to date classification systems for shocked rocks or sediments from the Earth, the Moon, Mars, and the asteroids. Here, we synthesize this latest classification system for feldspar-dominated rock types, with our results and the characteristic Raman data from shock stage S1 to S4 (see Table 5 and Fig. 13). It should be noted that all the Raman spectra stacked here were collected on intermediate plagioclase (~An_{50}) with EPMA-confirmed composition.

Our results for unshocked plagioclase reference materials (see Fig. 4) are consistent with previous studies (e.g., Sharma et al. 1983; Velde and Boyer 1985; Freeman et al. 2008). The most characteristic Raman features of feldspars are observed to be two or three Raman peaks between 460 and 515 cm^{-1}. The exact position and numbers of these peaks and the distance between the two diagnostic peaks vary with composition; intermediate plagioclase exhibits fewer peaks than the endmember (i.e., albite and anorthite) compositions, and the two distinct peak positions are not exactly at the midpoint between the Raman peaks of the crystalline endmembers. Unfortunately, there have not been any systematic experimental or theoretical studies demonstrating the influence of Na/Ca and Si/Al ratios upon peak positions and intensity in Raman spectra. Thus, while K-feldspar exsolution can be discerned from plagioclase by their Raman features, as shown in this study, the Raman peak positions are not sufficient for judging the composition of plagioclase solid solutions.

For unshocked to weakly shocked plagioclase (S1–S2 stages), our observations of the microscopic features in plagioclase include fractures, twins, and needle-like inclusions (Table 5). However, these features are not uniquely indicative of shock (French and Koeberl 2010). Among Raman spectra collected from deformed twins, the weakening of peaks below 350 cm^{-1} is consistent with a slight deformation of the crystal lattice. However, the intensity of peaks below 350 cm^{-1} is also orientation dependent on the lattice vibration. The intensity ratio of two signature peaks between 480 and 510 cm^{-1} also varies with different crystal orientation. Thus, while Raman spectra can be used to discern tiny microscopic features and deformed twins, they cannot be used as a decisive indicator of shocked plagioclase from these two stages.

For the S3 stage, the presence of PDFs is a prime microscopic criterion and is the most diagnostic criterion for moderately shocked rocks (Stöffler et al. 2018). As discussed above, while no PDFs were observed in plagioclase at the Mistastin Lake impact structure, planar features were used as a shock indicator. The increase in Raman peak width (FWHM) together with decrease of intensity and loss of weak peak suggests the distortion of the aluminosilicate framework.

From shock stage S1 to S3, as shown in Figs. 13A–13D, the FWHM of the two main diagnostic Raman peaks increase slightly from <20 to around 30 cm^{-1}. It is notable that we observed no significant peak shift within low to moderately shocked plagioclase. Our Raman results agree well with the previous studies on Martian meteorites (Fritz et al. 2005), and experimentally shocked plagioclase (Fritz et al. 2019). Combined, these studies show the same overall progression: as the shock level

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Table 5. Progressive stages of shock metamorphism in plagioclase.

Shock stage	Diagnostic shock effects (Stöffler et al. 2018)	Optical features in plagioclase (Mistastin Lake impact structure; this study)	Raman features (cm^{-1}) in plagioclase (Mistastin Lake impact structure; this study)
S1 Unshocked	Sharp optical extinction and irregular fractures	Fractures	185, 200, 285, 408, 480, 509, 568, 765, 795, 984, 1186, 1216 cm^{-1}
S2 Weakly shocked	Undulatory extinction	Undulatory extinction, deformed twins	Main diagnostic peaks show slight broadening, lattice modes peaks under 350 cm^{-1} slight weakening
S3 Moderately shocked	Partially isotropic and with PDFs and weak mosaicism	Partially isotropic, planar features	Broadening of the main peaks, loss of weak peaks
S4 Strongly shocked	Diaplectic glass	Diaplectic glass	Broad band near 510 cm^{-1} with shoulder near 580 cm^{-1}, broad peak centered around 1050 cm^{-1}, intensity varies
S5 Very strongly shocked	Normal glass	n/a	n/a

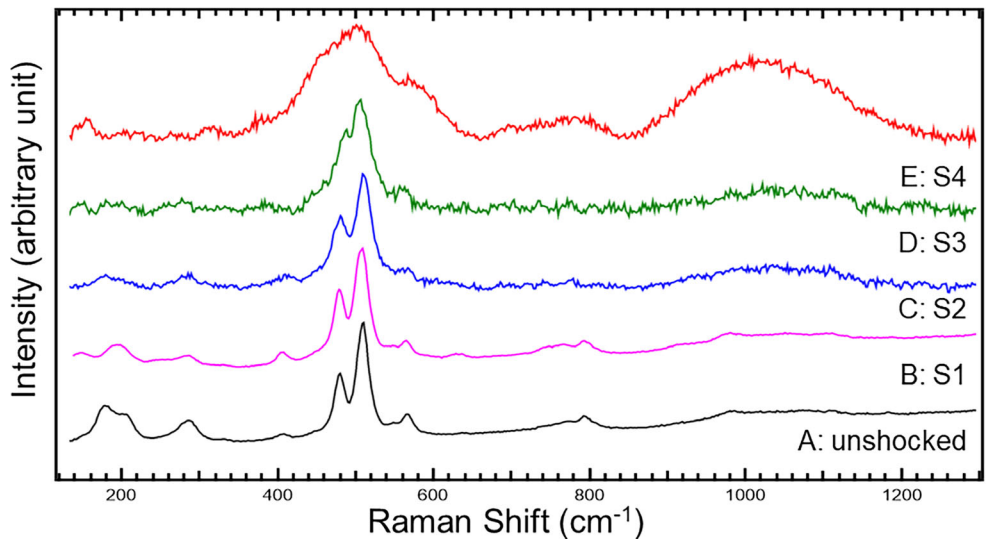


Fig. 13. Raman spectra of unshocked and shocked plagioclase (~An_{50} composition) from the Mistastin Lake impact structure. The background has been removed and the intensity has been normalized for better comparison. A) Stress- and strain-free plagioclase reference material from Labrador, Canada. B) Unshocked plagioclase from Mistastin Lake. C) Heavily deformed twins. D) Partial isotropic area. E) Diaplectic glass. (Color figure can be viewed at wileyonlinelibrary.com.)

increases, the Raman spectra show progressive and systematic changes in peak width (FWHM) and peak loss. For shock stage S4, diaplectic glass is the major indicator for this strongly shocked stage (Table 5). As discussed above, Raman features are sufficient to discern crystalline plagioclase and diaplectic glass due to the observable changes.

All the Raman features presented here demonstrate that Raman spectroscopy is a promising tool for discerning shock features at the microscopic scale and for distinguishing diaplectic glass and shocked crystalline plagioclase at various shock stages. However, there are several factors that need to be taken into consideration that otherwise may limit the full use of Raman in future terrestrial and planetary applications. First, beam size and resolution of the Raman system need to be considered. The laboratory Raman system used in this study has a beam size of 1–2 μm and the FWHM resolution is 0.3 cm^{-1}, whereas the Raman instrument in SuperCam onboard the Mars2020 Perseverance rover will have a beam size of ~5 mm (i.e., 5000 μm) and an FWHM resolution of <9 cm^{-1} (Wiens et al. 2019). The relatively large beam size and resolution make Raman spectra less efficient in discerning tiny microscopic features and low to moderately shocked stages in the field, but it is

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adequate for identifying diaplectic glass. Second, fluorescence and high Raman background are inevitable during Raman study of natural samples, and they can be induced by impurities within the sample. Adjusting laser power and collecting time can help to suppress fluorescence and Raman background. However, even when using laboratory Raman with a high confocal system, a rising background is still always observed in the Raman spectra, especially for moderately shocked plagioclase and diaplectic glass (as seen in Figs. 10 and 12). High background may mask the Raman features and make the results misleading. As such, when using Raman instruments in the field, it is critical to achieve decent signal-to-noise ratios in order to gain sufficient information from Raman spectra.

CONCLUSIONS

This study systematically examined plagioclase feldspar collected from a suite of impactites from the Mistastin Lake impact structure, Canada, using both optical microscopy, EPMA, and Raman spectroscopy. In summary:

1. A set of progressive shock features, including fractures, deformed twins, planar features, partially isotropic and diaplectic glass, are observed in shocked plagioclase from the Mistastin Lake impact structure. Undiagnostic features include fractures and deformed twins; diagnostic features include planar features and diaplectic glass. The Raman spectra collected from the microscopic features in samples from S1 to S3 shock stages show progressive changes, including band broadening, reduction of intensities, and loss of lattice mode bands, which can be attributed to microstructural deformation and loss of crystallinity.
2. Unlike crystalline plagioclase, the Raman spectra of diaplectic glass mainly consist of broad peaks in the high- (900–1200 cm^{-1}) and low-wave number (400–700 cm^{-1}) regions, assigned to stretching and bending vibrations of T-O-T (T = Si, A1) linkages and TO_4 groups, respectively. We suggest the intensity difference of Raman features observed in diaplectic glass is likely due to different *P–T–t* paths of different samples and pressure-induced disorder or heterogeneity. The presence of the orientated needle-like inclusions in both diaplectic glass and crystalline plagioclase in this study indicates that diaplectic glass is formed from pressure-induced amorphization and not melting.

Finally, plagioclase feldspar is one of the most abundant minerals on the surface of the Earth, the Moon, and Mars. Our results suggest that high precision Raman spectroscopy is a promising tool that is capable of discerning microscopic shock features and distinguishing between diaplectic glass and crystalline plagioclase. More systematic Raman studies on shocked feldspar with different compositions are needed to improve our understanding of shock metamorphism and planetary evolution.

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